(FS) 1. Continuous random variables HW

NO. 1

The continuous random variable X has cumulative distribution function F given by

$$\mathrm{F}(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{60}(16x - x^2) & 0 \leqslant x \leqslant 6, \\ 1 & x > 6. \end{cases}$$

a. Find the interquartile range of X. [4]

b. Find $\mathrm{E}(X^3)$. [4]

The random variable Y is such that $Y = \sqrt{X}$.

c. Find the probability density function of Y. [3]

A level FURTHER MATHEMATICS 2020/Winter/Paper4_1/Q6

Mark scheme

Question	Answer	Marks	Guidance
a	UQ: F(U) = 0.75: $u^2 - 16u + 45 = 0$	M1	Obtain quadratic equation for u or l
	$u = 8 - \sqrt{19}$ (= 3.64)	A1	Value of UQ
	LQ: $l^2 - 16l + 15 = 0$, l =1	A1	Value of LQ
	IQR = UQ – LQ = 7 − √19 = 2.64	A1 FT	
		4	

b	$f(x)=\begin{cases}\frac{1}{30}(8-x), & 0\leqslant x\leqslant 6\\ 0, & \text{otherwise}\end{cases}$	B1	May be implied, only need to see $f(x)=\frac{1}{30}(8-x)$
	Find $E(X^3)$ by integration: $\frac{1}{30}\int_0^6\left(8x^3-x^4\right)dx$ $=\left[2x^4-\frac{1}{5}x^5\right]_0^6$	M1 A1	Attempt integration (with PDF not CDF) limits not required Correct integrated expression, with limits
	= 34.56	A1	
		4	
c	$G(y) = \frac{1}{60}(16y^2 - y^4)$	M1	CDF for Y
	$g(y) = \frac{1}{60}(32y - 4y^3)$ for $0 \le y \le \sqrt{6}$	M1A1	Differentiate to find PDF for Y Correct $g(y)$ and correct range seen anywhere
		3	

Explanation

a.

UQ: F(U) = 0.75: $u^2 - 16u + 45 = 0$. M1 Obtain quadratic equation for u or l

$u = 8 - \sqrt{19}$ (= 3.64). A1 Value of UQ

LQ: $l^2 - 16l + 15 = 0$, l =1. A1Value of LQ

IQR = UQ – LQ = $7 - \sqrt{19} = 2.64$. A1FT

b.

$$f(x)=\begin{cases}\frac{1}{30}(8-x), & 0\leqslant x\leqslant 6\\ 0, & \text{otherwise}\end{cases}$$

B1 May be implied, only need to see $\mathrm{f}(x)=\frac{1}{30}(8-x)$

Find $\mathrm{E}(X^3)$ by integration:

$$\frac{1}{30}\int_0^6\left(8x^3-x^4\right)\mathrm{d}x$$

$$=\left[2x^4-\frac{1}{5}x^5\right]_0^6$$

M1 A1Attempt integration (with PDF not CDF)limits not required Correct integrated expression, with limits

= 34.56. A1

c.

$G(\mathrm{y}) = \frac{1}{60}(16\mathrm{y}^2 - \mathrm{y}^4)$. M1 CDF for Y

$g(\mathrm{y}) = \frac{1}{60}(32\mathrm{y} - 4\mathrm{y}^3)$ for $0 \le \mathrm{y} \le \sqrt{6}$. M1A1 Differentiate to find PDF for Y Correct $g(\mathrm{y})$ and correct range seen anywhere

NO. 2

The continuous random variable X has probability density function f given by

$$\mathrm{f}(x)=\begin{cases}\frac{1}{5}x & 0\leqslant x<2,\\ \frac{2}{15}(5-x) & 2\leqslant x\leqslant 5,\\ 0 & \text{otherwise.}\end{cases}$$

a. Find the cumulative distribution function of X. [3]

b. Find the median value of X. [2]

c. Find $\mathrm{E}(X^2)$. [2]

d. Find $\mathrm{P}(1 \le X \le 3$. [2]

Mark scheme

Question	Answer	Marks
a	$\mathrm{F}(x)=\begin{cases}\dfrac{x^2}{10}, & 0\leqslant x<2\\ \dfrac{1}{15}\left(10x-x^2\right)-\dfrac{2}{3}, & 2\leqslant x\leqslant 5\end{cases}$	M1A1
	= 0 for x < 0 and = 1 for x > 5	A1
		3
b	$\mathrm{F}(m)$ = ½ so $2m^2 - 20m + 35 = 0$	M1
	$m=5-\dfrac{1}{2}\sqrt{30}=2.26$	A1
		2
c	$\mathrm{E}(X^2)=\int_0^2 x^2.\dfrac{x}{5}\mathrm{d}x+\int_2^5\dfrac{2}{15}(5-x).x^2\mathrm{d}x=\left[\dfrac{x^4}{20}\right]+\left[\dfrac{2}{15}\left(\dfrac{5}{3}x^3-\dfrac{x^4}{4}\right)\right]$	M1
	= 6.5	A1
		2
d	$\mathrm{F}(3)-\mathrm{F}(1)$	M1
	$=\dfrac{11}{15}-\dfrac{1}{10}=\dfrac{19}{30}$	A1
	Alternative method for 3(d)	

	$\int_1^2 \frac{x}{5}\,dx + \int_2^3 \frac{2}{15}(5-x)\,dx = \left[\frac{x^2}{10}\right] + \left[\frac{2}{15}\left(5x - \frac{x^2}{2}\right)\right]$	M1
	$= \frac{3}{10} + \frac{1}{3} = \frac{19}{30}$	A1
		2

Explanation

a.

$$F(x) = \begin{cases} \frac{x^2}{10}, & 0 \leqslant x < 2 \\ \frac{1}{15}(10x - x^2) - \frac{2}{3}, & 2 \leqslant x \leqslant 5 \end{cases}$$ M1 A1

= 0 for $x < 0$ and = 1 for $x > 5$. A1

b.

$\frac{1}{15}(10m - m^2) - \frac{2}{3} = \frac{1}{2}$

$5m - \frac{1}{2}m^2 = \frac{35}{4}$

$F(m) = ½$ so $2m^2 - 20m + 35 = 0$. M1

$m = 5 - \frac{1}{2}\sqrt{30} = 2.26$ A1

c.

$$E(X^2) = \int_0^2 x^2 . \frac{x}{5}\,dx + \int_2^5 \frac{2}{15}(5-x).x^2\,dx = \left[\frac{x^4}{20}\right] + \left[\frac{2}{15}\left(\frac{5}{3}x^3 - \frac{x^4}{4}\right)\right]$$ M1

= 6.5. A1

d.

F(3) – F(1) M1

$= \frac{11}{15} - \frac{1}{10} = \frac{19}{30}$ A1

Alternative method for 3(d)

$\int_1^2 \frac{x}{5}\,dx + \int_2^3 \frac{2}{15}(5-x)\,dx = \left[\frac{x^2}{10}\right] + \left[\frac{2}{15}\left(5x - \frac{x^2}{2}\right)\right]$ **M1**

$= \frac{3}{10} + \frac{1}{3} = \frac{19}{30}$ **A1**

NO. 3

The continuous random variable X has distribution function given by

$$F(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{90}(x^2 + x^4) & 0 \leqslant x \leqslant 3, \\ 1 & x > 3. \end{cases}$$

The random variable Y is defined by $Y = X^2$.

i. Find the probability density function of Y. [4]

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ii. Find the mean value of Y. [2]

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A level FURTHER MATHEMATICS 2018/Winter/Paper2_2/Q7

Mark scheme

Question	Answer	Marks	Guidance
7.i	*EITHER:* G(y) [= P(Y < y) = P(X^2 < y) = P(X < $y^{1/2}$) = F($y^{1/2}$)] = (1/90)(y + y^2)	**M1A1**	Find or state G(y) for $0 \leqslant x \leqslant 3$ from $Y = X^2$ (allow < or $\leqslant$ throughout)

	OR: Use $x = y^{1/2}$ to find $f(x) = (1/90)(2x + 4x^3) = (1/90)(2y^{1/2} + 4y^{3/2})$ *and* $dx/dy = 1 / 2y^{1/2}$	**(M1A1)**	Find $f(x)$ *and* dx/dy for use in $g(y) = f(x) \times \lvert dx/dy \rvert$
	$g(y)$ [= $G'(y)$] = $(1/90)(1 + 2y)$	**A1**	Find $g(y)$ in simplified form
	for $0 \leqslant y \leqslant 9$ [$g(y) = 0$ otherwise]	**A1**	State corresponding range of y at any stage
		4	
ii	$E(Y) = (1/90)\int(y + 2y^2)dy$	**M1**	Find mean of Y from $\int y\, g(y)\, dy$
	$= (1/90)[\frac{1}{2} y^2 + \frac{2}{3} y^3]_0^9 = 117/20$ *or* 5.85	**A1**	
		2	

Explanation

7.i

EITHER:

$G(y)$ [= $P(Y < y) = P(X^2 < y)$

$= P(X < y^{1/2}) = F(y^{1/2})$] $= (1/90)(y + y^2)$. M1 A1 Find or state $G(y)$ for $0 \leqslant x \leqslant 3$ from $Y = X^2$(allow < or $\leqslant$ throughout)

OR:

Use $x = y^{1/2}$ to find

$f(x) = (1/90)(2x + 4x^3) = (1/90)(2y^{1/2} + 4y^{3/2})$ *and* $dx/dy = 1 / 2y^{1/2}$ M1 A1 Find $f(x)$ *and* dx/dy for use in $g(y) = f(x) \times \lvert dx/dy \rvert$

$g(y)$ [= $G'(y)$] = $(1/90)(1 + 2y)$. A1 Find $g(y)$ in simplified form

for $0 \leqslant y \leqslant 9$ [$g(y) = 0$ otherwise]. A1 State corresponding range of y at any stage

ii.

$E(Y) = (1/90)\int(y + 2y^2)dy$. M1 Find mean of Y from $\int y\, g(y)\, dy$

$= (1/90)[\frac{1}{2} y^2 + \frac{2}{3} y^3]_0^9 = 117/20$ *or* 5.85. A1

NO. 4

The continuous random variable X has probability density function f given by

$$\mathrm{f}(x) = \begin{cases} \frac{1}{6}\left(x^{-\frac{1}{3}} - x^{-\frac{2}{3}}\right) & 1 \leqslant x \leqslant 27, \\ 0 & \text{otherwise}. \end{cases}$$

a. Find the cumulative distribution function of X. [3]

The random variable Y is defined by $Y = X^{\frac{1}{3}}$.

b. Find the probability density function of Y. [3]

c. Find the exact value of the median of Y. [2]

A level FURTHER MATHEMATICS 2023/Summer/Paper4_3/Q1

Mark scheme

Question	Answer	Marks	Guidance
a	$\mathrm{F}(x)=\frac{1}{6}\left[\frac{3}{2}x^{\frac{2}{3}}-3x^{\frac{1}{3}}\right]+c$	M1	Integrate f(x), +c not required, at least one power correct.
	$\frac{1}{4}x^{\frac{2}{3}}-\frac{1}{2}x^{\frac{1}{3}}+\frac{1}{4}$	A1	AEF
	Correct ranges including $1 \leqslant x \leqslant 27$ associated with their $\mathrm{F}(x)$ $\mathrm{F}(x) = 0$ for $x < 1$ and $\mathrm{F}(x) = 1$ for $x > 27$	A1	No gaps.
		3	
b	$\mathrm{G}(y)=\frac{1}{4}\left(y^2-2y+1\right)$	M1	Using $y=x^{\frac{1}{3}}$ in *their* $\mathrm{F}(x)$.
	$\mathrm{g}(y)=\frac{1}{2}\left(y-1\right)$	A1	Correct, AEF, 0 otherwise not required.
	For $1 \leqslant \mathrm{y} \leqslant 3$	B1	Seen anywhere, correct variable.
		3	

c	$\frac{1}{4}\left(y^2-2y+1\right)=\frac{1}{2}, \quad \left(y-1\right)^2=2$ OE	M1	Equate their G(y) to ½ and solve to find y.
	$\left[y=\right] \quad 1+\sqrt{2}$	A1	Only.
		2	

NO. 5

The continuous random variable X has cumulative distribution function F given by

$$\mathrm{F}(x)=\begin{cases}0 & x<0,\\ 1-\frac{1}{144}(12-x)^2 & 0\leqslant x\leqslant 12,\\ 1 & x>12.\end{cases}$$

a. Find the upper quartile of X. [2]

b. Find Var(X^2). [5]

The random variable Y is given by $Y = \sqrt{X}$.

c. Find the probability density function of Y. [3]

A level FURTHER MATHEMATICS 2022/Winter/Paper4_1/Q5

Mark scheme

Question	Answer	Marks	Guidance
a	$\mathrm{F}(U) = 0.75$, so $1-\frac{1}{144}(12-U)^2 = \frac{3}{4}$	M1	
	$(12 - U)^2 = 36$, $12 - U = \pm 6$, $U = \pm 6$ only	A1	
		2	
b	$[\mathrm{f}(x) =] \quad \frac{1}{72}(12-x)$	B1	Differentiate given PDF.
	$\mathrm{E}(X^2) = \int_0^{12} \frac{1}{72}(12x^2 - x^3)\,dx = \frac{1}{72}\left(4x^3 - \frac{1}{4}x^4\right)$ [= 24]	M1	Integrate $x^2f(x)$ with their $f(x)$.

	$E\left(X^4\right)=\int_0^{12}\frac{1}{72}\left(12x^4-x^5\right)dx=\frac{1}{72}\left(\frac{12}{5}x^5-\frac{1}{6}x^6\right)$ [= 1382.4]	M1	Integrate $x^4f(x)$ with their $f(x)$.
	Var(X^2) = E(X^4) − E^2(X^2) [=1382.4 − 24^2]	M1	Using correct formula with numerical values possibly unsimplified.
	1382.4 − 576 = 806(.4)	A1	
		5	
c	$\mathrm{G}(y)=\begin{cases}0 & y<0,\\ 1-\frac{1}{144}\left(12-y^2\right)^2 & 0\leqslant y\leqslant\sqrt{12},\\ 1 & y>\sqrt{12}.\end{cases}$	M1	Change the variable: $1-\frac{1}{144}\left(12-y^2\right)^2$ Allow domain unchanged at this stage.
	$\frac{1}{36}\left(12y-y^3\right)$	M1	Differentiate their G(y).
	$g\left(y\right)=\begin{cases}\frac{1}{36}\left(12y-y^3\right) & 0\leqslant y\leqslant\sqrt{12}\\ 0 & \text{otherwise}\end{cases}$	A1	Fully correct.
		3	

NO. 6

The continuous random variable X has cumulative distribution function F given by

$$\mathrm{F}(x)=\begin{cases}0 & x<-1,\\ \frac{1}{2}(1+x)^2 & -1\leqslant x\leqslant 0,\\ 1-\frac{1}{2}(1-x)^2 & 0<x\leqslant 1,\\ 1 & x>1.\end{cases}$$

a. Find the probability density function of X. [2]

b. Find $\mathrm{P}\left(-\frac{1}{2} \leqslant X \leqslant \frac{1}{2}\right)$. [2]

c. Find $\mathrm{E}(X^2)$. [2]

d. Find Var(X^2). [2]

A level FURTHER MATHEMATICS 2021/Winter/Paper4_1/Q2

Mark scheme

Question	Answer	Marks	Guidance
a	$\mathrm{f}(x)=\begin{cases}1+x & -1 \leqslant x \leqslant 0, \\ 1-x & 0<x \leqslant 1, \\ 0 & \text{otherwise.}\end{cases}$	M1	Differentiation attempted.
		A1	All correct, including 0 otherwise.
		2	
b	$\mathrm{P}\left(-\frac{1}{2} \leqslant X \leqslant \frac{1}{2}\right)=\mathrm{F}\left(\frac{1}{2}\right)-\mathrm{F}\left(-\frac{1}{2}\right)=\frac{7}{8}-\frac{1}{8}$	M1	Can do by integration with correct limits.
	$\frac{3}{4}$	A1	
		2	
c	$\int_{-1}^{0} x^2(1+x)\,dx+\int_{0}^{1} x^2(1-x)\,dx$ $\left[\frac{1}{3}x^3+\frac{1}{4}x^4\right]+\left[\frac{1}{3}x^3-\frac{1}{4}x^4\right]$	M1	
	$\frac{1}{12}+\frac{1}{12}=\frac{1}{6}$	A1	

		2	
d	$\int_{-1}^{0} x^4(1+x)\,dx + \int_{0}^{1} x^4(1-x)\,dx - \left(\frac{1}{6}\right)^2$ $\left[\frac{1}{5}x^5 + \frac{1}{6}x^6\right] + \left[\frac{1}{5}x^5 - \frac{1}{6}x^6\right] - \frac{1}{36}$	M1	Complete method with their mean squared explicit.
	$\frac{1}{15} - \frac{1}{36} = \frac{7}{180}$	A1	
		2	